INTEL AI FOR FUTURE WORKFORCE BOOTCAMP 2026 TECHNICAL DOCUMENTATION SERIES

AI Models, Agent Functions and Execution Capabilities

Comprehensive Architectural & Model Evaluation Dossier for VOID AI

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1. Model-to-Function Mapping Table

Component	Model / Technology	Primary Function	Execution Mode	Status
Local Ultra-Lite Brain	`nano-brain:latest` (SmolLM2-360M-Instruct)	Ultra-low-latency local conversational response, intent extraction, coding assistance, and reasoning under resource caps.	Local CPU (AVX2 FMA, <150MB RAM) via Ollama	PRODUCTION
Local Multi-Modal Brain	`lfm3-omni:2b` (Liquid AI 1.7B Q4_K_M)	Multi-modal local agentic reasoning, multi-turn tool calling, structured planning, and image comprehension.	Local Ollama Daemon (GPU/VRAM Standby)	PRODUCTION
Cloud Decider Bridge	OpenRouter / Perplexity `pplx-decider-flash-v1.3:free`	Optional high-complexity DAG planning, fallback when local models are offloaded, and deep domain synthesis.	Cloud API (HTTPS REST with Secret Masking)	PRODUCTION
Acoustic Wake Recognizer	Vosk Kaldi FST (vosk-model-small-en-us-0.15)	Continuous real-time open-grammar listening for wake triggers (`"hey void"`, `"void"`) with <120ms latency.	Local CPU Subprocess (PipeWire Stream)	PRODUCTION
Speech-to-Text (STT)	Faster-Whisper Base int8 (AVX2 4-thread CPU)	High-accuracy spoken utterance transcription with dynamic Silero VAD silence trimming (~350ms latency).	Local CPU int8 Quantized	PRODUCTION
Neural Speech Synthesis	Piper Neural TTS (vits-piper-en_US-ryan-medium)	Natural, high-fidelity 22.05 kHz voice generation (160+ wpm at 1.06x speed) via native PipeWire streaming.	Local ONNX Runtime (Native PipeWire pw-play)	PRODUCTION
Multi-Agent Orchestrator	VOID AI DAG Core (Supervisor & Planner)	Decomposes natural language goals into Directed Acyclic Graphs with dependency arrays and role assignments.	Local Python Async Engine (`agentic/`)	PRODUCTION
Tool Hub & DBus	105 VOID AI Native Tools + KWin Wayland DBus	Direct operating system interaction: window management, process inspection, filesystem manipulation, and documents.	Native Linux Subprocesses & DBus Calls	PRODUCTION

2. Architectural Taxonomy: LLM vs. Agent vs. Orchestration vs. Executable Tool

To avoid conceptual ambiguity, VOID AI maintains a strict architectural distinction across four tiers:

- **Foundation LLM (The Cognitive Engine):** The neural language model (`nano-brain`, `lfm3-omni:2b`, or cloud provider) acts purely as an inference predictor. It interprets language, classifies intents, and predicts structured responses. It does *not* possess execution authority.
- **Agent (The Role-Bounded Actor):** An autonomous software entity encapsulating a specific role (e.g., Developer Agent, Linux Intelligence Agent, Verifier Agent). Each agent has a defined system prompt, constrained tool access, state tracking, and specialized decision boundaries.
- **Orchestration Framework (The DAG Coordinator):** The Supervisor and Planner agents in `agentic/orchestrator.py` that manage workflow state machines (`WorkflowLifecycleState`), schedule task execution order based on `depends_on` dependencies, and route results between agents.
- **Executable Tool (The Actuator):** Sandboxed, deterministic Python/Shell functions registered in `ToolRegistry` (lines 20-100 of `agentic/tools.py`). Tools interact with the physical OS: reading `/proc`, scaffolding directories, executing `pytest`, or making DBus calls.

3. The Autonomous Deliberation & Planning Cycle

1. **Goal Ingestion:** The user gives a natural-language command (voice or text). Vosk wakes the system, Faster-Whisper transcribes the audio, and phonetic normalization standardizes vocabulary.
2. **Intent Classification:** The `SupervisorAgent` inspects the request. Direct desktop controls (volume, window toggle, timers) route immediately to the sub-5ms fast-path. Multi-faceted complex requests activate the agentic pipeline.
3. **DAG Formulation:** The `PlannerAgent` analyzes the objective, breaks it down into discrete sub-tasks, assigns specialized agent roles, maps input/output dependencies (`depends_on`), and assigns security levels (Level 1 to Level 4).
4. **Task Execution:** The orchestrator executes tasks respecting dependency order. Execution agents invoke sandboxed tools and capture `ToolResult` outputs including exit codes, execution duration, and generated artifacts.
5. **Independent Verification Gate:** The `VerifierAgent` independently evaluates ground truth: verifying file existence, inspecting stdout/stderr for errors, checking return codes, or executing unit test suites.
6. **Self-Healing Recovery:** If the verifier detects failure, the `RecoveryAgent` inspects error traces, formulates an alternative strategy, and applies a bounded retry (max 2 attempts).
7. **Completion & Memory Commit:** On successful verification, the Supervisor commits the execution trace to the 7-day rolling memory pool (`memory/weekly_episodes.jsonl`), records an immutable audit log, and synthesizes a natural spoken response via Piper TTS.

4. Local Model Lifecycle & Memory Preservation

VOID AI addresses the resource constraints of edge personal computers through dynamic model lifecycle management:

- **Ultra-Lite nano-brain Residency:** Consumes only 98MB to 140MB RAM. Responds in ~210ms to ~350ms, enabling real-time local chat with near-zero CPU and memory impact.

- **On-Demand lfm3-omni:2b Loading:** When multi-modal tool calling is needed, Ollama loads the 1.7B GGUF weights into APU/GPU memory.
- **Instant Standby Offloading:** Upon voice command (*"Hey VOID, stop model"* or *"offload model"*), the system terminates the `llama-server` process, immediately freeing ~2.0 GB of RAM in under 20ms while seamlessly switching to Cloud AI fallback.

5. Online vs. Offline Capabilities Matrix

Subsystem / Capability	Offline (Local)	Online (Cloud)	Execution Mechanism
Acoustic Wake-Word Detection	100% OFFLINE	N/A	Vosk Kaldi FST on local CPU
Speech-to-Text Transcription	100% OFFLINE	Optional Cloud STT	Faster-Whisper Base int8 (AVX2)
Sub-5ms Window & Media Controls	100% OFFLINE	N/A	KWin DBus / WirePlumber
Linux System Vitals & Diagnostics	100% OFFLINE	N/A	Direct `/proc`, `df`, socket inspection
Autonomous Software Scaffolding	100% OFFLINE	Enhanced via Cloud	Python `DeveloperAgent` & `pytest`
Publication Document Drafting	100% OFFLINE	Enhanced via Cloud	python-docx & Headless Chromium
Deep Live Web Research	Offline Snippet Cache	LIVE WEB SEARCH	DuckDuckGo Multi-Source Scraper
Speech Generation (TTS)	100% OFFLINE	N/A	Piper Neural ONNX via `pw-play`